Daniel Yang

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EDUCATION

University of Washington - M.S. Electrical and Computer Engineering | Expected Winter 2026

Purdue University - B.S. Mechanical Engineering, Minor in Electrical and Computer Engineering | Dec. 2022 | GPA: 3.6/4.0

RESEARCH EXPERIENCE

University of Washington | Graduate Research Assistant, HiPER Lab | Seattle, WA | Apr. 2026 - Present

- 6.78 MHz dynamic wireless power transfer: Developed and integrated a moving-receiver WPT prototype with a Class Phi-2 GaN inverter, high-Q copper coils, rectifier and motorized gantry; demonstrated approximately 120 W output and 80% peak DC-to-DC efficiency under tested conditions.
- Designed, fabricated and characterized resonant copper coils; measured impedance, inductance and quality factor using a vector network analyzer (VNA) and tuned series capacitance to 6.78 MHz.
- Performed Ansys HFSS electromagnetic simulations of coil coupling, aluminum-frame eddy currents and coil-to-frame spacing; evaluated mutual inductance and magnetic fields to inform mechanical layout and parasitic-loss tradeoffs.
- Designed and built a belt-driven linear-gantry test platform; implemented STM32 stepper-motor control and trapezoidal motion profiles for dynamic transfer experiments.
- Self-resonant WPT coil research: Created HFSS models and geometry sensitivity studies to analyze inductance, equivalent series resistance (ESR), Q factor, current density and conductor losses at 6.78 MHz; cross-checked inductance using Modified Wheeler and Current Sheet analytical approximations. Optimization remains ongoing.

Purdue University | Nanotechnology Research Assistant | West Lafayette, IN | Jan. 2021 - May 2021

- Assisted with research, synthesis and testing of hybrid CNT-ZIF-MoS2 nanostructure supercapacitors for energy-storage applications.

PROFESSIONAL EXPERIENCE

Boeing Commercial Airplanes | Mechanical/Hydraulics Steering Systems Engineer | Everett, WA | Jul. 2023 - Aug. 2025

- Served as responsible engineer for landing gear hydraulic and electrohydraulic actuators, metering valves and servo valves on Boeing 737/767 platforms; supported component qualification, fleet safety, supplier quality and production engineering.
- Authored revisions to the 777-200F / 777-9 System Safety Analysis (SSA); conducted qualitative FMEA and Functional Hazard Assessment (FHA), assessed failure propagation, redundancy and aircraft-level hazards, and obtained approval for revised certification documentation under 14 CFR requirements.
- Initiated a root-cause investigation of an in-service 777 steering anomaly; analyzed flight and qualification-test data, proposed test scenarios, secured leadership budget approval and supported a one-week flight-test campaign; identified the failure mechanism and recommended corrective action.
- Led KC-46A aerial-refueling receptacle door actuator requalification as prime project engineer; coordinated fatigue/loading requirements, supplier test plans, structural-stress report review, schedule and technical stakeholders; secured agreement on revised qualification documentation.
- Investigated 737/767 fleet steering anomalies and supplier defects using flight-data analysis, hydraulic fault isolation and root-cause analysis; communicated safety findings to review boards and engineering leadership.
- Resolved manufacturing nonconformances through GD&T, engineering drawing interpretation, tolerance studies and cross-functional stress/quality reviews; supported engineering dispositions and production continuity.

PROFESSIONAL EXPERIENCE (CONTINUED)

Boeing Commercial Airplanes | Mechanical/Hydraulics Steering Systems Engineer Intern | Everett, WA | Jun. 2022 - Aug. 2022

- Built a MATLAB four-bar-linkage kinematics and quasi-static load model for a redesigned 777 main landing gear steering actuator; combined linkage geometry, NASA tire mechanics relationships and flight-test data to estimate actuator forces versus steering angle.
- Created 3D mechanism visualization and loading plots to support structural evaluation of the steering actuator.

Honda Aircraft Company | Hydro/Landing Gear Systems Engineer Intern | Greensboro, NC | May 2021 - Aug. 2021

- Designed and delivered a fabricated nose/main landing gear retraction-extension test fixture using CATIA V5, FEA, MATLAB static calculations, engineering drawings and GD&T; coordinated BOM procurement and machining.
- Developed a ground-spoiler fault isolation manual using hydraulic/electrical schematics and FMEA-based troubleshooting logic; procedures were used by production mechanics.

GE Appliances | Washer Systems Center of Excellence Co-op | Louisville, KY | Jun. 2020 - Aug. 2020

- Analyzed washer acoustic recordings using Audacity spectrograms and Excel to support competitive noise benchmarking against consumer review ratings.

GE Appliances | Supply Chain Area Business Leader Co-op | Louisville, KY | Aug. 2019 - Dec. 2019

- Supported high-volume washer assembly operations across two production lines; used defect and first-pass-yield data to coordinate inspection improvements. Team FPY increased from approximately 85% to 92% during the rotation.

SELECTED ENGINEERING PROJECTS

Rolls-Royce Sponsored Senior Design | Automated Dry-Ice Vat Cooling System | Purdue University

- Led a four-person team developing an automated dry-ice dispensing prototype; designed the ESP32 control architecture, schematic and firmware with thermocouple SPI sensing, motor drivers, relays and LCD interface; integrated thermal logic with mechanical augers and SolidWorks/FEA frame design.

Power Electronics Design and Control | University of Washington

- Designed and tested DC-DC boost converters using circuit simulation, small-signal/averaged modeling, transfer-function analysis and closed-loop controller design.

TECHNICAL SKILLS

Power electronics / electromagnetics: Resonant WPT, 6.78 MHz ISM systems, GaN Class Phi-2 inverter integration, DC-DC converters, coil coupling, HFSS, VNA measurement, impedance, Q factor, ESR, eddy currents, skin/proximity effects

Mechanical / systems engineering: CATIA V5, SolidWorks, Fusion 360, FEA, GD&T, DFM, hydraulic/electrohydraulic systems, root-cause analysis, FMEA, FHA, SSA, qualification testing, 14 CFR certification support, supplier coordination

Programming / embedded: MATLAB, Simulink, Python, C, STM32, ESP32, Arduino, LTspice, SystemVerilog

Languages: English and Japanese (fluent)